

IIT Kharagpur

Department of Mathematics

Foundations of AI & ML

(LTP- 3-0-0,CRD- 3, 40hrs)

Prerequisites:

Probability and Statistics (MA20205)

Theory:

1. Introduction to graphs (2 lectures)
2. BFS/DFS (2 lectures)
3. Best first search, depth bounded search, A* search, uniform cost search (3 lectures)
4. Algorithmic Game theory, chomp, NIM (2 lectures)
5. Propositional logic (2 lectures)
6. Decision trees, minimax, alpha-beta pruning (3 lectures)
7. Introduction to learning (2 lectures)
8. Clustering: k means, agglomerative hierarchical clustering, density-based spatial clustering of applications with noise (3 lectures)
9. Regression (LM, GLM) (3 lectures)
10. Gradient descent (2 lectures)
11. Perceptron (2 lectures)
12. Deep neural networks (2 lectures)
13. Back propagation (2 lectures)
14. Monte Carlo tree search(2 lectures)
15. Introduction to Evolutionary computation(2 lectures)
16. Genetic algorithms (3 lectures)
17. Swarm algorithms(3 lectures)

References:

1. Russel, S. (2010). Artificial Intelligence: A Modern Approach. Stuart J. Russell and Peter Norvig.
2. Brabazon, A., O'Neill, M., & McGarraghy, S. (2015). *Natural computing algorithms* (Vol. 554). Berlin: Springer.
3. Bishop, C. M. (2006). *Pattern recognition and machine learning*. Springer.
4. James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). *An introduction to statistical learning* (Vol. 112, p. 18). New York: springer.
5. Shalev-Shwartz, S., & Ben-David, S. (2014). *Understanding machine learning: From theory to algorithms*. Cambridge university press
6. Murphy, K. P. (2012). *Machine learning: a probabilistic perspective*. MIT press
7. Hastie, T., Tibshirani, R., Friedman, J., & Franklin, J. (2005). The elements of statistical learning: data mining, inference and prediction. *The Mathematical Intelligencer*, 27(2), 83-85.
8. Vapnik, V. (2013). *The nature of statistical learning theory*. Springer science & business media.

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